

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

BIOLOGY

9744/02

Paper 2 Structured Questions

23 August 2022

Candidates answer on the Question Paper.
No Additional Materials are required

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, Biology class and registration number at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/10
2	/10
3	/10
4	/10
5	/10
6	/10
7	/10
8	/10
9	/10
10	/10
Total	/100

This document consists of **28** printed pages.

- 1 Fig. 1.1 shows the reproductive cycle of human herpesvirus (HHV).

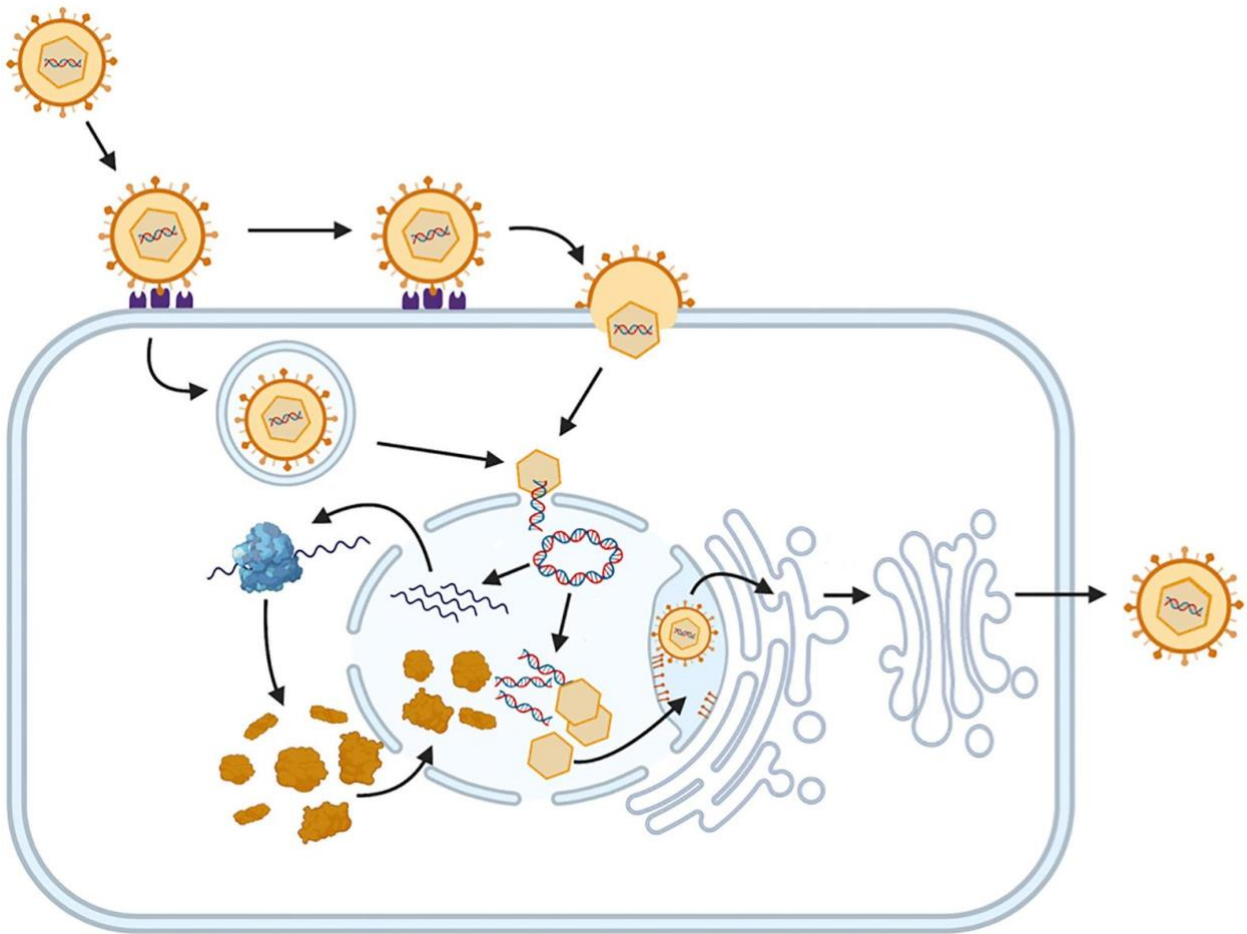


Fig. 1.1

- (a) With reference to Fig. 1.1, describe the structure of HHV.

[3]

- (b) Name **two** membranous organelles that are shown in Fig. 1.1 and outline their roles in the reproductive cycle of HHV.

1

2

[4]

- (c) Discuss how viruses such as HHV challenge the cell theory and concepts of what is considered living.

[3]

[Total: 10]

- 2 (a) Fig. 2.1 shows the structure of the amino acid cysteine.

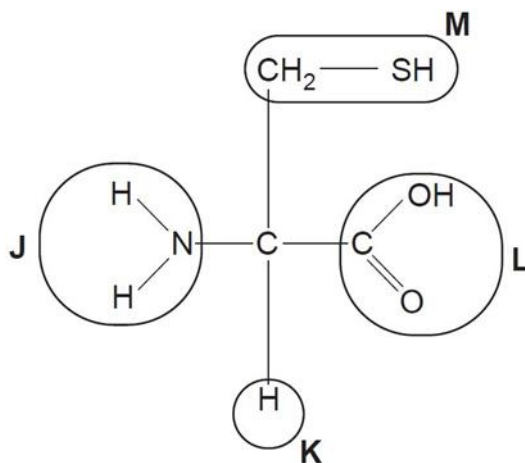


Fig. 2.1

- (i) Complete the table by selecting the letter, J, K, L or M, that represents each of the following groups in cysteine.

group	letter
carboxyl	
amino	

[1]

- (ii) The primary structure of a protein consists of a chain of amino acids.

With reference to Fig. 2.1, draw how two cysteines would be joined together in forming the primary structure of a protein. Label and name the bond(s) formed.

[3]

- (iii) Describe the role of group **M** in the protein structure.

[1]

- (b) Pepsin is a protease found in the stomach.

Fig. 2.2 shows the effect of varying substrate concentration on pepsin activity.

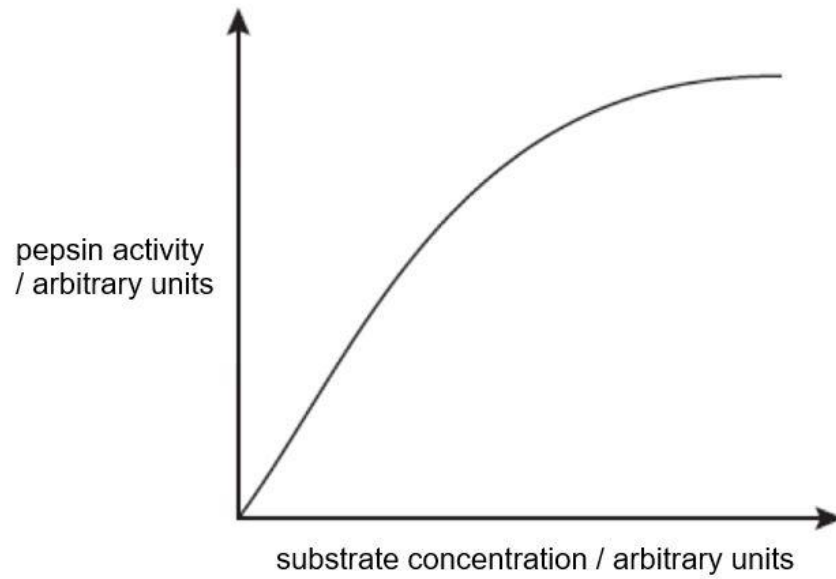


Fig. 2.2

- (i) Pepstatin is known to be a non-competitive inhibitor of pepsin.

Explain the effects of pepstatin on pepsin.

[3]

- (ii) On Fig. 2.2, draw a graph to show how the activity of pepsin would vary with its substrate concentration in the presence of pepstatin. Annotate the maximal velocity as ' $V_{\max}$ (inhibited)' on the graph that you have drawn.

[2]

[Total: 10]

- 3 Fig. 3.1 shows the percentage of rat cells undergoing DNA replication. Some cells contained a protein called cyclin D and some cells did not. All cells were in early interphase at time 0.

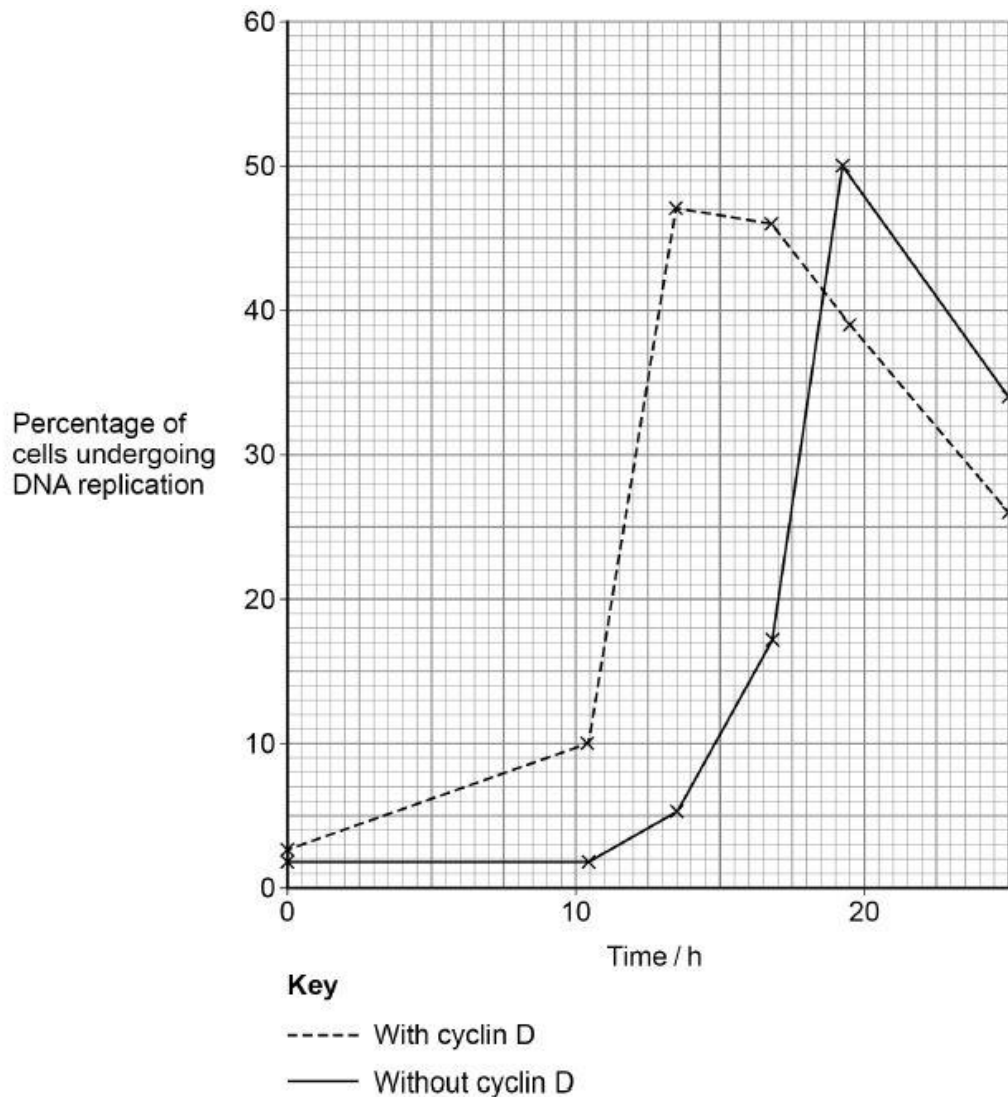


Fig. 3.1

- (a) Name the stage in early interphase when DNA replication occurs.

[1]

- (b) With reference to Fig. 3.1, describe the effect of cyclin D on cells.

[2]

- (c)** Cyclin D is known to bind specifically to a cyclin-dependent kinase (Cdk), which then acts on DNA polymerase.

Suggest how cyclin D may cause a change in the activity of DNA polymerase.

[2]

- (d)** Describe the role of DNA polymerase in DNA replication.

[2]

- (e) Fig. 3.2 shows a linear chromosome undergoing DNA replication.

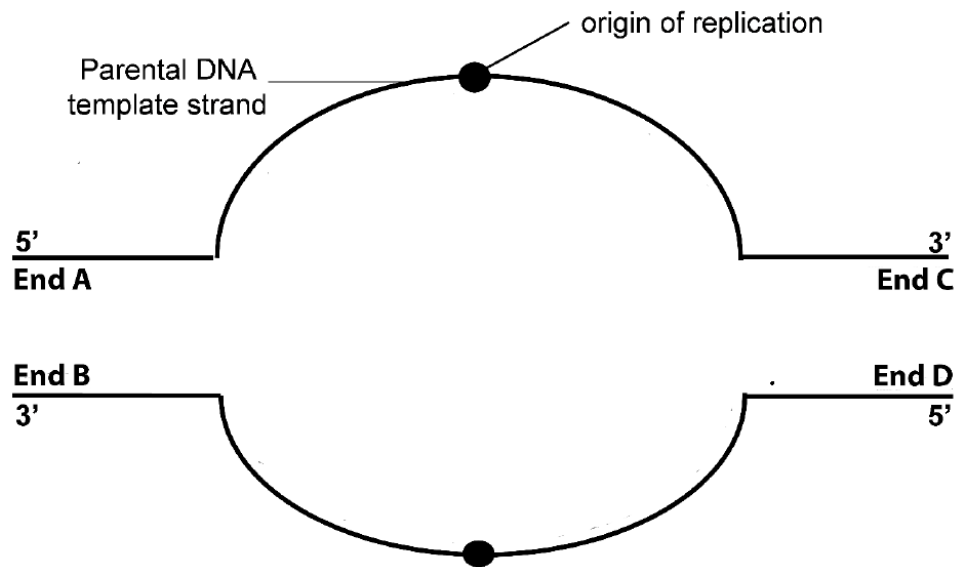


Fig. 3.2

- (i) On Fig. 3.2, draw the direction of DNA replication for **one** leading ($\longrightarrow$) strand and **one** lagging ($\cdots\longrightarrow$) strand.

[2]

- (ii) Ends **A**, **B**, **C** and **D** represent the ends of the newly synthesised daughter DNA strands.

State which ends will experience the end-replication problem after the first round of DNA replication has been completed.

[1]

[Total: 10]

- 4 Fig. 4.1 shows the pathways that can be activated by glucagon to increase the blood glucose level when it is bound to its receptor on the liver cell surface.

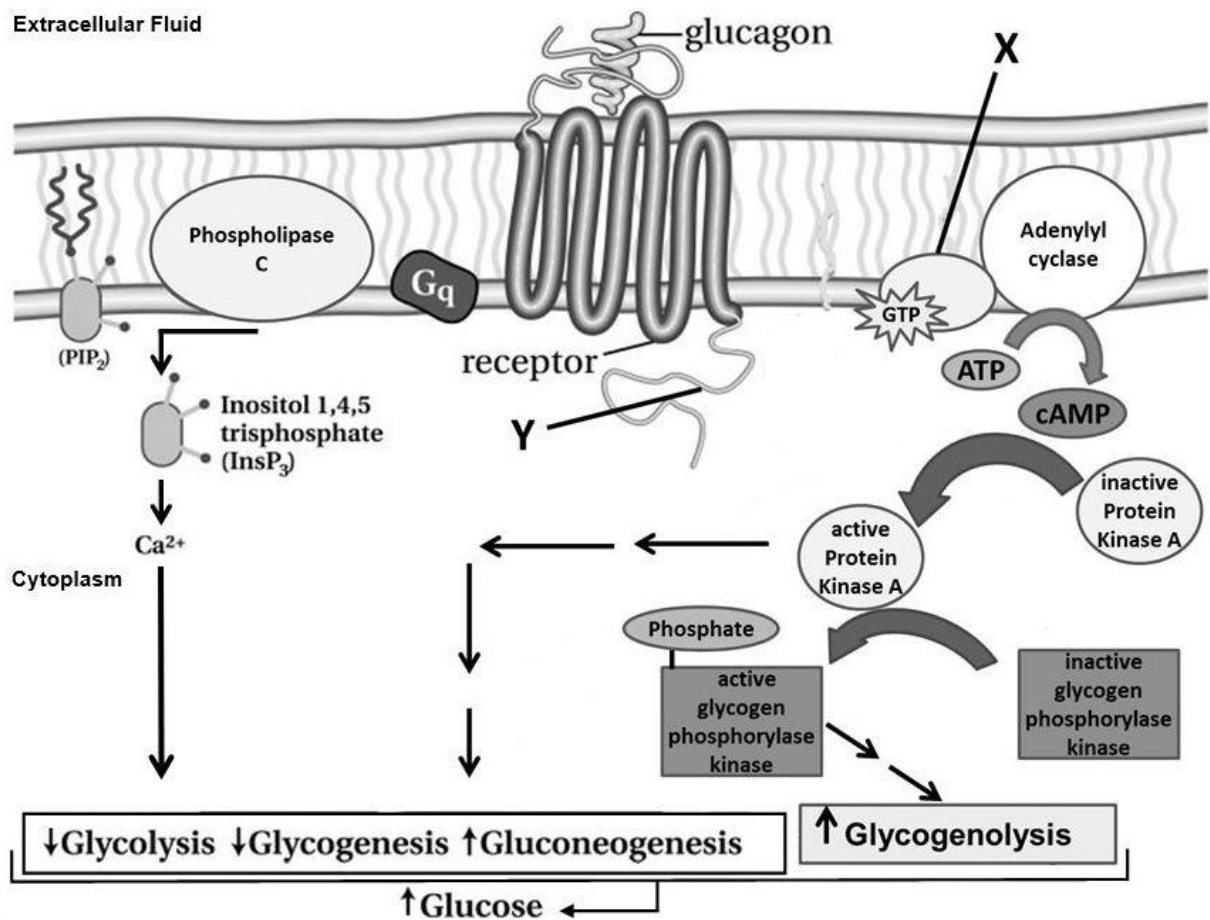


Fig. 4.1

- (a) Name molecule X in Fig. 4.1.

[1]

- (b) With reference to Fig. 4.1,

- (i) suggest the function of region Y of the glucagon receptor.

[1]

- (ii) describe how glucagon increases the blood glucose level when it is bound to its receptor on the effector cell surface.

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[5]

- (c) A group of liver cells has been found to contain a mutation in a gene coding for phosphatase **W**. Phosphatase **W** acts on a particular substrate in the glucagon signalling pathway to prevent glycogenolysis.

With reference to Fig. 4.1, suggest the identity of the substrate of phosphatase **W** and explain how glycogenolysis is prevented.

.....

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.....

.....

.....

[3]

[Total: 10]

- 5 Induced pluripotent stem cells (iPSCs) have become a model system for studying tissue development and related human diseases, holding great promise for cell therapy. Recent studies found genetic abnormalities in iPSCs, which may harbour risk of tumorigenesis.

Fig. 5.1 shows the karyotype of an abnormal iPS cell line that is maintained in a research lab.

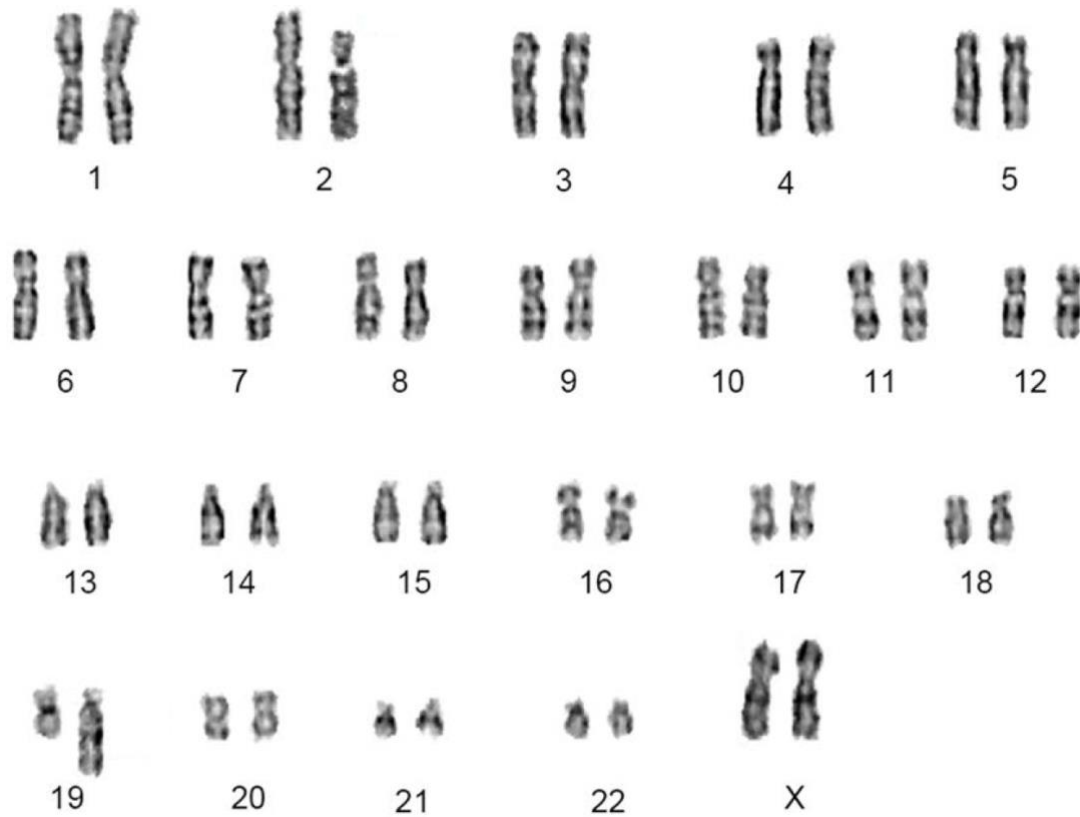


Fig. 5.1

- (a) Describe the genetic abnormality shown in Fig. 5.1.

[2]

Fig 5.2 shows the lineage scheme of the somatic cell that was transformed into iPSC.

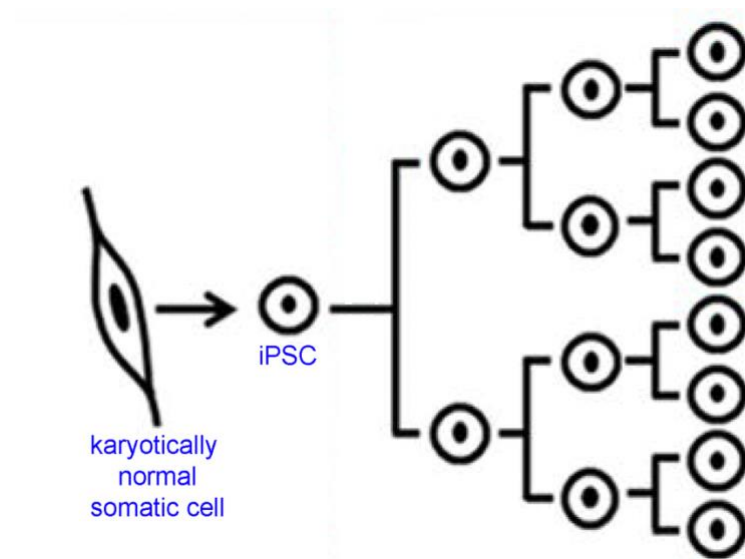


Fig. 5.2

(b) The genetic abnormality was found in approximately 53.3% of the progeny cells. All genetically abnormal iPSCs in the cell line carried the same karyotype.

(i) On Fig. 5.2, indicate and label where the genetic abnormality occurred. [1]

(ii) Outline the events that resulted in occurrence of genetic abnormality in 53.3% of the progeny.

[4]

- (c)** It is hypothesised that as iPSCs undergo a rapid increase in the rate of cell division, it leads to an increase in energy demand. This causes a significant increase in the level of reactive oxygen species, resulting in oxidative DNA damage.

(i) Name another cause of DNA damage.

..... [1]

(ii) Suggest how DNA damage in the iPSCs led to the karyotype shown in Fig. 5.1.

.....
.....
.....
..... [2]

[Total: 10]

- 6** Tomato varieties vary in many characters such as stature (tall versus dwarf) and skin structure (smooth versus peach skin). Plants from F_1 generation which were tall and had smooth fruits were self-pollinated.

In the F_2 generation, 101 plants were tall and had smooth fruits, 56 plants were tall and had peach fruits, 52 plants were dwarf and had smooth fruits, and none were dwarf and had peach fruits.

- (a)** Draw a genetic diagram to explain the distribution of plants in the F_2 generation.

Use the symbols **A**, **a** and **B**, **b** to represent the alleles.

[4]

- (b)** Carry out chi-squared test to support your answer in **(a)**.

- (i)** State the null hypothesis.

[1]

(ii) Complete the table.

F ₂ phenotype	observed numbers (O)	expected numbers (E)	$\frac{(O - E)^2}{E}$
tall smooth	101		
tall peach	56		
dwarf smooth	52		
dwarf peach	0		
$\chi^2 = \sum \frac{(O - E)^2}{E}$			

[3]

(iii) Critical values of this chi-squared distribution are shown in Table 6.1.

Table 6.1

degrees of freedom	probability								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
2	0.02	0.10	0.21	0.58	1.39	2.77	4.61	5.99	9.21
3	0.12	0.35	0.58	1.21	2.37	4.11	6.25	7.81	11.34
4	0.30	0.71	1.06	1.92	3.36	5.39	7.78	9.49	13.28

degrees of freedom = number of classes - 1

State the conclusions that may be drawn from **(b)(ii)**.

[2]

[Total: 10]

7 Fig. 7.1 shows the reduction of NAD during cellular respiration.

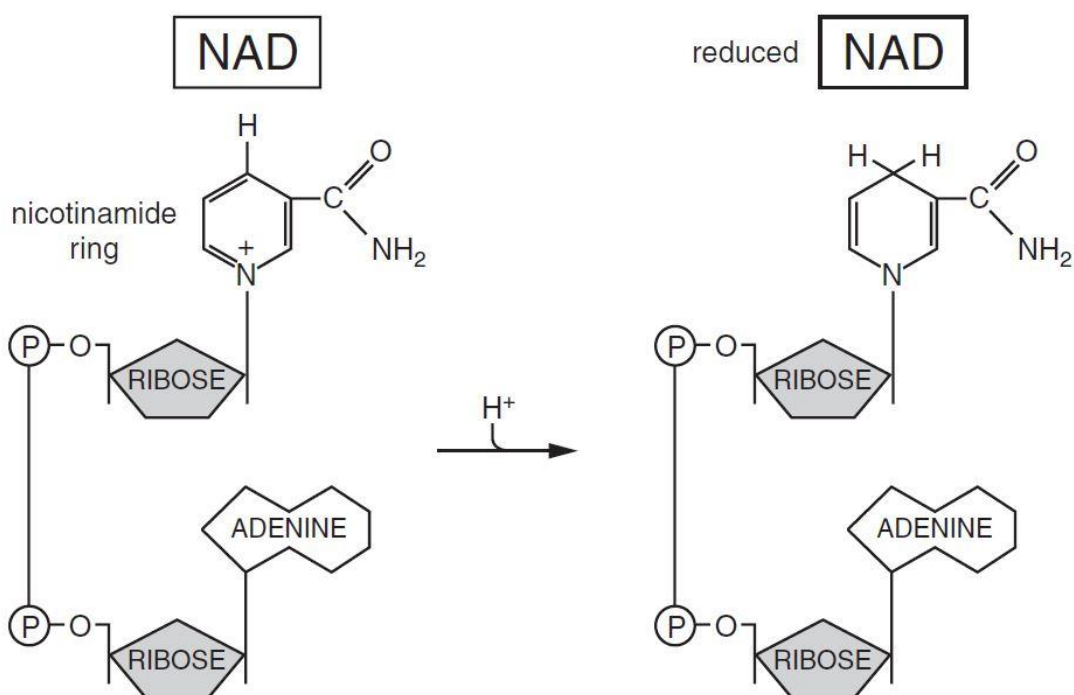


Fig. 7.1

(a) State **two** locations in the eukaryotic cell where NAD is reduced.

[2]

(b) Outline the role of NAD in cellular respiration.

[3]

- (c) Explain why NAD cannot be regenerated from reduced NAD in mitochondria in the absence of oxygen.

[3]

- (d) Yeast can respire aerobically and anaerobically. When there is insufficient oxygen, yeast cells switch from aerobic to anaerobic respiration. This results in a significant increase in the rate of glucose uptake and glycolysis in the yeast cells.

Explain why the rate of glycolysis increases significantly when yeast cells switch from aerobic to anaerobic respiration.

[2]

[Total: 10]

- 8 Fig. 8.1 shows the development of myeloid and lymphoid stem cells from hematopoietic stem cell.

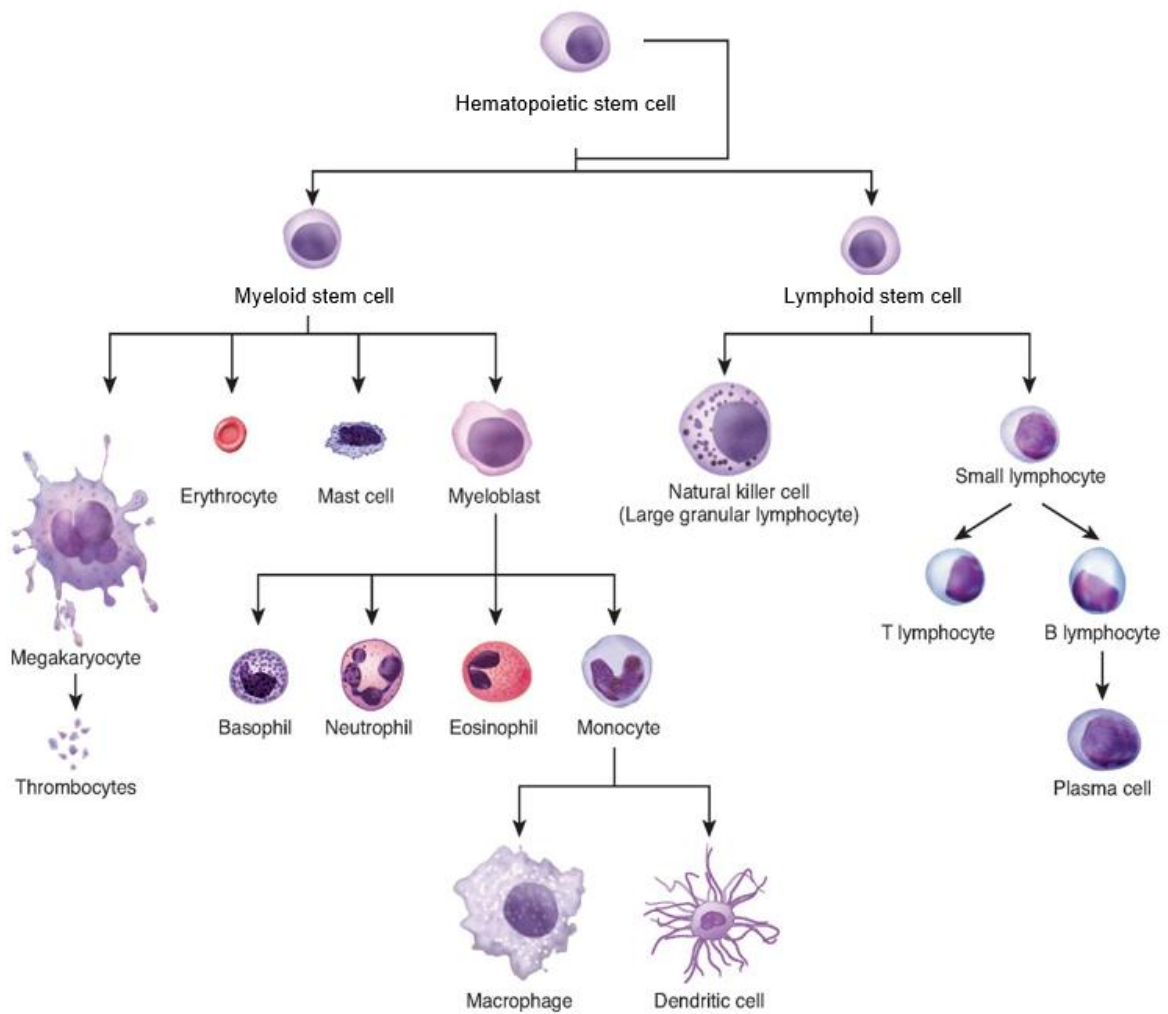


Fig. 8.1

- (a) With reference to Fig. 8.1, describe the features and functions of lymphoid stem cells.

[3]

Natural killer cell deficiency (NKD) is a primary immunodeficiency where the main defect lies in NK cells. One possible cause of NKD is the *GATA2* mutation.

GATA2 is a transcription factor that regulates the proliferation, differentiation and survival of hematopoietic stem cells. GATA2 has a zinc finger domain that allows for DNA-protein interaction.

Fig. 8.2 shows the zinc finger interacting with a segment of DNA.



Fig. 8.2

(b) Explain how the zinc finger domain of GATA2 can bind to the DNA double helix.

[3]

- (c) NK cells show strong cytolytic functions against physiologically stressed cells such as tumour cells and virus-infected cells.

Suggest and explain how GATA2 mutation may lead to increased incidence of cancer.

[4]

[Total :10]

- 9 Scientists at the European Molecular Biology Laboratory in Germany studied the activity of genes involved in making the sleep hormone, melatonin, and the melatonin activity in the marine worm, *Platynereis dumerilii*.

Fig. 9.1 shows a two-day old larva of *P. dumerilii*. The larva has locomotor-ciliated cells that regulates vertical swimming.

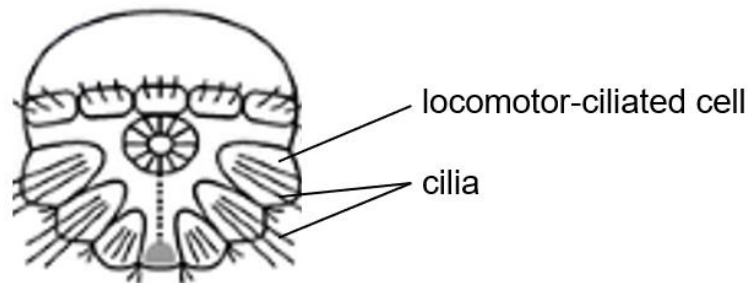


Fig. 9.1

During light hours, melatonin production is inhibited. During dark hours, melatonin is produced. Melatonin induces rhythmic burst firing by motor neurons to release acetylcholine (ACh) that targets locomotor-ciliated cells. Ciliary arrests were found to be triggered by ACh transmission.

Fig. 9.2 shows the interaction between light-dark hours, melatonin production and regulation of vertical swimming.

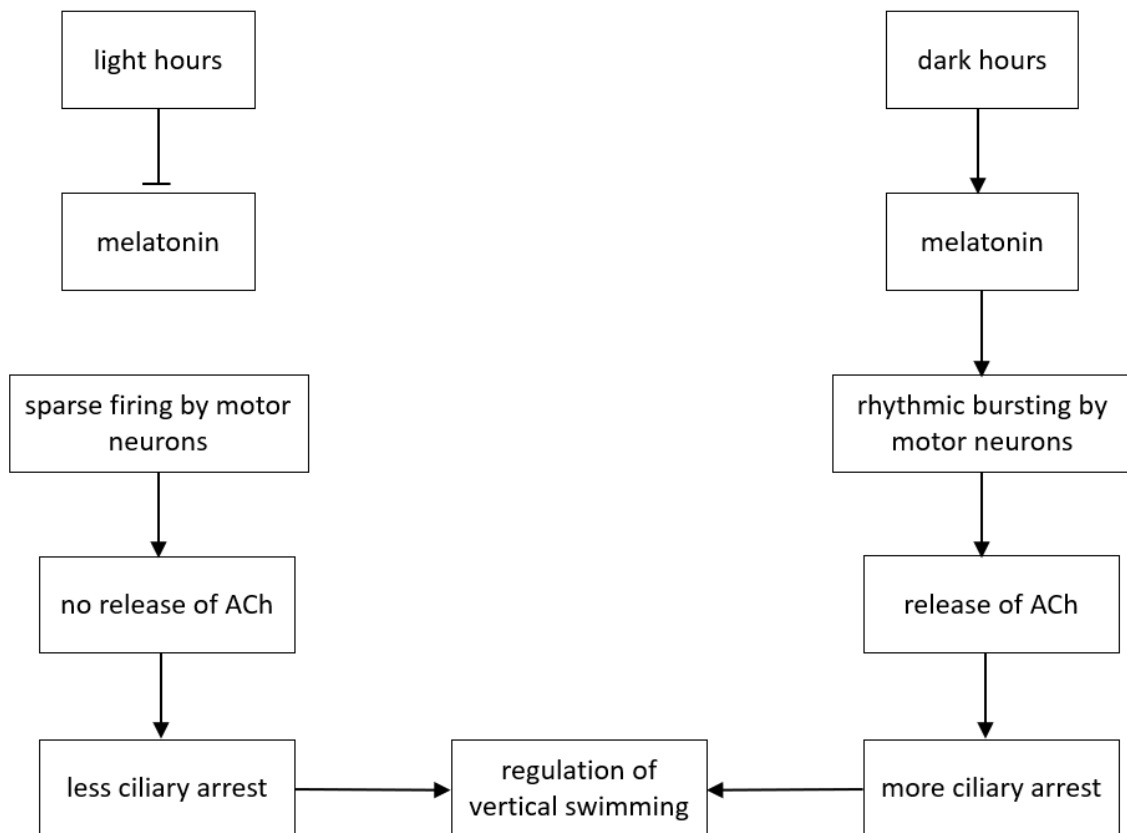


Fig. 9.2

- (a) With reference to Fig. 9.2, suggest and explain whether the larva will be near the surface or bottom of the water body when there is light.

[3]

- (b) Suggest and explain **one** evolutionary advantage of the reaction to the sleep hormone for the marine worm, *P. dumerilii*.

[2]

(c) Fig. 9.3 shows the biochemical pathway of melatonin synthesis.

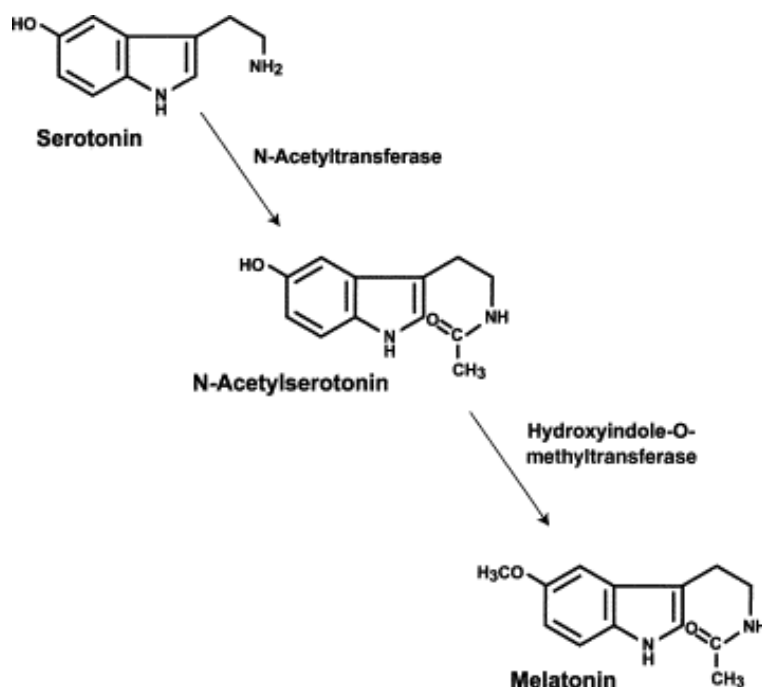


Fig. 9.3

The process to synthesise melatonin from serotonin requires two enzymes, N-acetyltransferase and hydroxyindole-O-methyltransferase. Recessive alleles that are not able to code for either of the enzymes have been preserved in the population of *P. dumerilii*.

Explain how the recessive alleles may be preserved in the population of *P. dumerilii*.

[2]

- (d) *P. dumerilii* are considered living fossils as fossils of this organism have been found as far back as the Cambrian era (about 500 million years ago). *P. dumerilii* are thought to be the last common ancestor of almost all living animals, including worms, flies and humans. A population of *P. dumerilii* developed photoreceptor cells and ability to produce and react to melatonin.

Explain why the population is the smallest unit that can evolve.

[3]

[Total :10]

- 10** Dengue viruses are spread to people through the bites of infected *Aedes aegypti*. These mosquitoes typically lay eggs near standing water in containers that hold water.
- (a)** Using the space below, draw a labelled diagram of the life cycle of the *A. aegypti* mosquito and indicate the duration of each stage.

[2]

Fig. 10.1 shows the outbreak prevalence of dengue virus and flushing prevalence in Singapore for a typical year. Flushing refers to a large quantity of water passing over a surface.

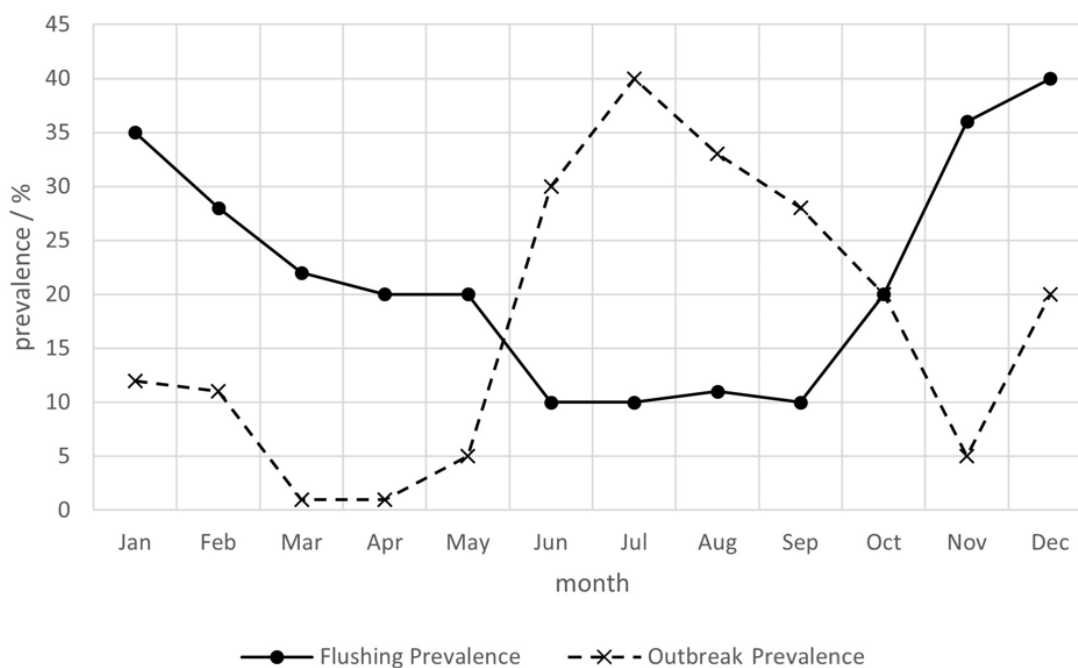


Fig. 10.1

(b) With reference to Fig. 10.1,

(i) describe how flushing prevalence changes over the months in a typical year.

[2]

(ii) suggest and explain how flushing prevalence affects outbreak prevalence of dengue virus.

[2]

(c) On Fig. 10.1, draw a graph to show how global warming can affect outbreak prevalence. [1]

(d) Describe and explain what is shown by the graph that you have drawn in **(c)**.

[2]

(e) Suggest why an individual can contract dengue more than once.

[1]

[Total: 10]